

Day 9: Docker Storage, Networks & Maven Builds

Volumes, Network Drivers, and Maven Lifecycle Automation

Module 1: Docker Volumes & Networking

By default, Docker containers are ephemeral. Any data created inside a container is lost when the container is deleted. Persistent storage and inter-container communication require volumes and custom network drivers.

Persistent Storage: Docker Volumes

- **Bind Mounts:** Maps an absolute directory path on the host system filesystem directly into the container directory. Useful for local code development.
- **Named Volumes:** Managed entirely by Docker in a dedicated storage area. Recommended for production databases and backups.

Docker Networking Drivers

Driver	Isolation Boundary	Primary Use Case
bridge	Default isolated host network	Multi-container apps on a single host (app server + database)
host	Shares host network namespace directly	High-performance applications to eliminate network translation overhead
overlay	Multi-host network communication	Docker Swarm or Kubernetes clusters spanning multiple VM hosts
none	Completely isolated network namespace	Offline batch processing, file processing, and secure calculators

Module 2: Apache Maven Build Automation

Apache Maven is a project management and build automation tool primarily used for Java applications. It manages dependencies, build life cycles, and documentation using a single configuration file: pom.xml.

Maven Build Lifecycle Phases

- **validate:** Verifies that the project structure is correct and all required configurations are present.
- **compile:** Translates source Java files (.java) into compiled bytecode classes (.class).
- **test:** Runs unit tests using frameworks like JUnit. If tests fail, the build terminates.
- **package:** Compresses compiled classes and resources into distributable archives (e.g. JAR, WAR).
- **install:** Copies the generated package into the local Maven repository (~/.m2/repository) for other local projects.
- **deploy:** Pushes the final package to a remote repository (like Nexus) for sharing with other

developers.